

The use case descriptions provide a solid foundation for the system's functionality, covering essential aspects like authentication, profile management, and security. However, some areas require improvement to ensure a seamless user experience and prevent potential usability issues. This feedback highlights both the strengths and areas that need refinement.

Positive feedback

The Sign-Up process is well-structured, ensuring smooth onboarding with strong input validation, including unique usernames, valid emails, and strong passwords. Email verification enhances security and prevents spam, while proactive exception handling improves user experience. The Login system efficiently authenticates users, providing seamless access with a "Stay Logged In" option that balances convenience and security. Error handling for incorrect credentials and password resets ensures accessibility, and the security measures account for failed login attempts. The Update Profile functionality allows users full control over their information, including privacy settings and avatars. The verification step for email changes ensures accuracy, and real-time validation improves usability. Consideration for re-authentication when making sensitive changes is a strong security measure, further enhancing user confidence in the system.

Areas for improvement

The current use case descriptions lack a Quit Use Case, which is essential for handling game exits. There is no defined behavior for confirming an exit, saving progress, or managing active matches. Without this, players might lose progress, and opponents in multiplayer sessions might not receive proper notifications. Implementing a structured Quit Use Case would allow seamless session saving and better user experience when exiting the game.

The Create Account GUI contains a typographical error in Scenario Step 2, where *"takes goes"* should be corrected to *"goes."* Additionally, the placement of *"Are you human?"* verification in Scenario Step 8 is incorrect—it should be conducted before email confirmation rather than after account creation. To improve clarity and security, the password strength requirements should be explicitly defined, including minimum length and special characters. The system should also specify what happens if the username or email is already taken before submission. Furthermore, clear guidelines on error message formats, whether inline, pop-up, or modal, should be included.

The Login GUI contains an incorrect description, mistakenly referring to the process as *"Create an Account"* instead of *"Log in to an account."* This mislabeling could confuse users. Enhancements should include defining retry limits for incorrect login attempts and specifying consequences, such as account lockout or CAPTCHA verification. Session timeout policies should be clearly stated, indicating if users will be logged out due to inactivity. Addressing these issues will greatly enhance the clarity, security, and user experience of the GUI use cases.

Overall, the use case descriptions demonstrate a strong focus on usability and security. While authentication and profile management are well-handled, addressing the missing Quit Use Case will improve the overall experience by ensuring a structured exit process. Implementing these refinements will enhance both user satisfaction and system reliability.